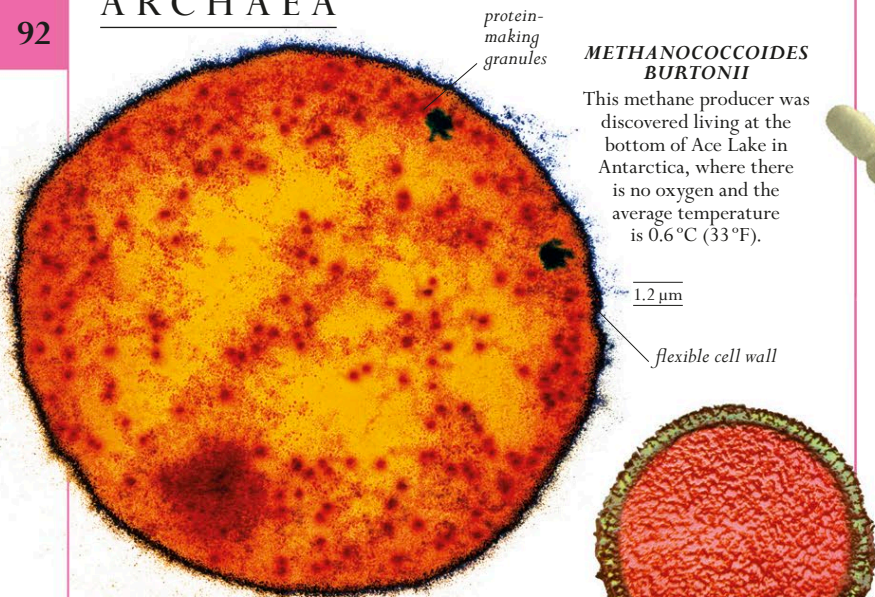


ARCHAEA



METHANOCOCCOIDES BURTONII

This methane producer was discovered living at the bottom of Ace Lake in Antarctica, where there is no oxygen and the average temperature is 0.6°C (33°F).

1.2 μm

flexible cell wall

0.5–15 μm

STAPHYLOTHERMUS MARINUS

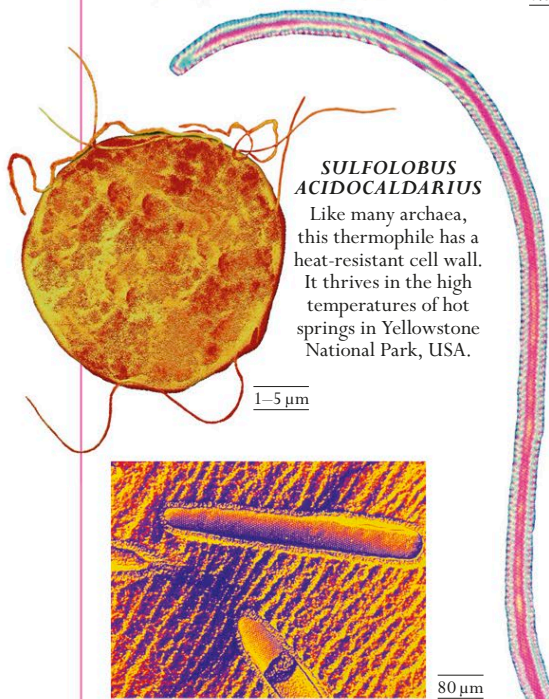
Discovered in a hydrothermal vent on the sea floor, this species grows best at 85–92°C (184–198°F). It forms grape-like clusters, and can grow comparatively large.



1 μm

DESULFUROCOCCUS MOBILIS

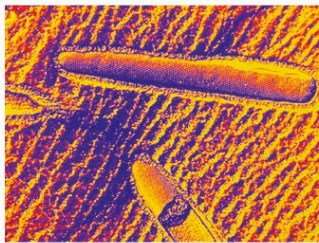
This anaerobe thrives where oxygen is limited or absent and uses sulfur-containing compounds in place of oxygen. An extreme thermophile, it grows best at 85°C (184°F).



SULFOLOBUS ACIDOCALDARIUS

Like many archaea, this thermophile has a heat-resistant cell wall. It thrives in the high temperatures of hot springs in Yellowstone National Park, USA.

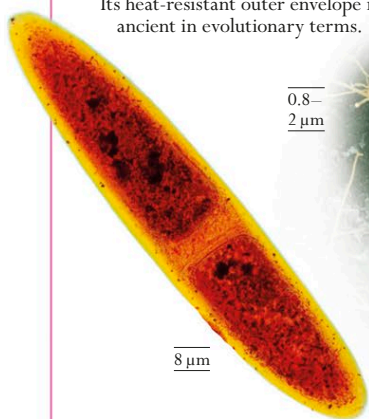
1–5 μm



80 μm

THERMOPROTEUS TENAX

This species forms rod-shaped cells of varying length but constant diameter. Its heat-resistant outer envelope is ancient in evolutionary terms.



8 μm

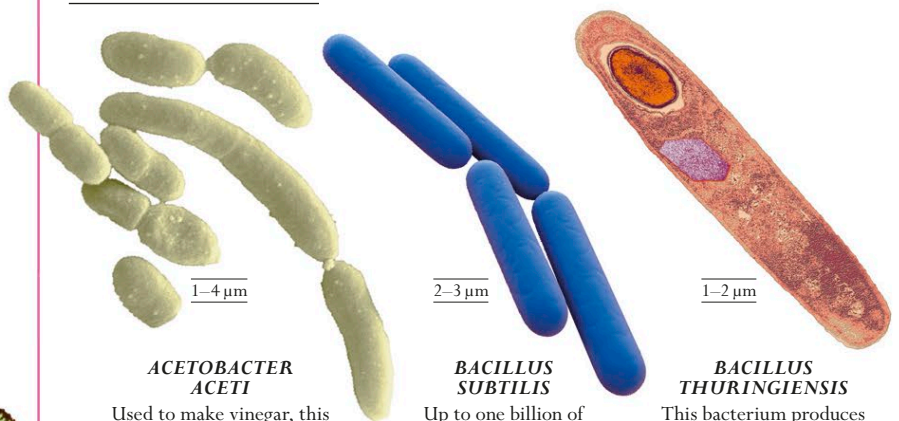
METHANOSPIRILLUM HUNGATEI

Discovered in human sewage, this archaeon generates large amounts of methane during sewage disposal. Each cell is confined within a hollow sheath.

PYROCOCCUS FURIOSUS

Pyrococcus means “fireberry”, reflecting this archaeon’s shape and its tolerance of extreme temperatures: it grows best at 100°C (212°F).

BACTERIA



ACETOBACTER ACETI

Used to make vinegar, this species is also a common contaminant wherever alcohol is brewed, causing discoloration and souring of beer in particular.

BACILLUS SUBTILIS

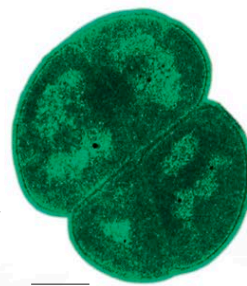
Up to one billion of these bacteria can be found in a single gram of soil. When food is scarce or the environment becomes harsh it can live as an inactive spore.

BACILLUS THURINGIENSIS

This bacterium produces toxic crystals insoluble in the human gut but lethal to insects, and is therefore commercially marketed as an insecticide.

DEINOCOCCUS RADIODURANS

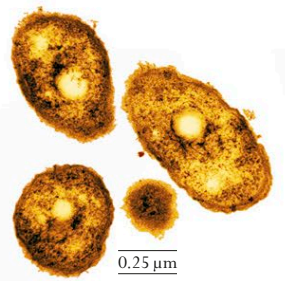
This species is known as the world’s toughest bacterium. It was discovered in meat that had been exposed to doses of radiation in an experiment.



1.62 μm

BORDETELLA PERTUSSIS

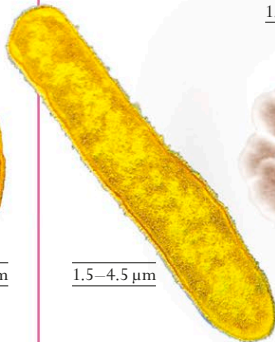
This species causes whooping cough, or pertussis, a respiratory infection marked by severe, spasmodic coughing episodes, and breakdown of blood cells.



0.25 μm

ESCHERICHIA COLI

An organism much observed and used by scientists, the rod-shaped *E. coli* lives harmlessly in the gut, although virulent strains are a major cause of food poisoning.



1.5–4.5 μm

BACTEROIDES FRAGILIS

Forming a large part of the human gut’s flora, this bacterium is usually harmless to its host. But it can also cause disease, invading tissues to form pus-filled abscesses.

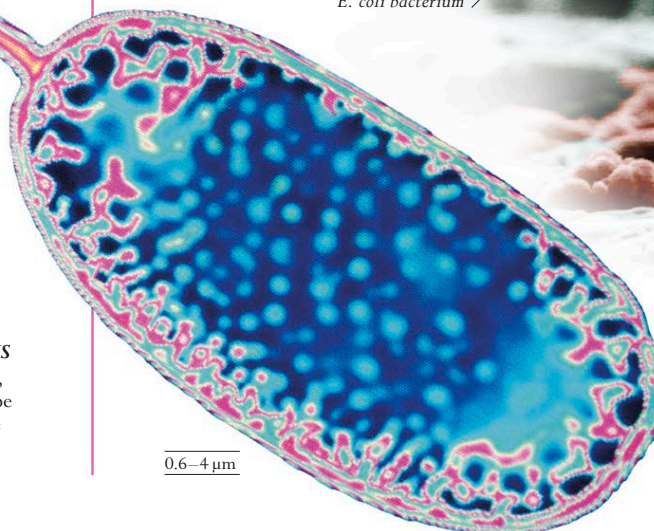


1–3 μm

individual rod-shaped *E. coli* bacterium

NITROBACTER SP.

This genus of bacteria plays a role in the nitrogen cycle by oxidizing nitrites into nitrates. This helps purify water and is central to fertilizing the oceans and the soil.



0.6–4 μm